1 Basics
- Taylor: $f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\dots$
  - $f(\boldsymbol{a})+\frac{\partial f(x)}{\partial x}\Big|_{\boldsymbol a}-\frac{1}{2}(x-\boldsymbol a)^\top\left(\frac{\partial^2 f(x)}{\partial x\partial x^\top}\right)\Big|_{\boldsymbol a}(x-\boldsymbol a)$
  - Power series of exp.: $\exp(x):=\sum_{k=0}^\infty \frac{x^k}{k!}$
- Entropy: $H(X)=\mathbb{E}_X[-\log\mathbb{P}(X=x)]$
- Diverg.: $D_{KL}(P\|Q)=\sum_{x\in\mathcal X}P(x)\log\left(\frac{P(x)}{Q(x)}\right)\geq 0$
- $1-z\le\exp(-z)$
- Cauchy-Schwarz: $|\mathbb E[X,Y]|^2\le\mathbb E[X^2]\mathbb E[Y^2]$
- Jensen, $f(X)$ convex: $f(\mathbb E[X])\le\mathbb E[f(X)]$
- Tot exp: $\mathbb E[X]=\mathbb E[\mathbb E[X|Y]]$

1.1 Probability / Statistics
- Gaussian: $\mathcal N(x|\mu,\sigma^2)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac12\frac{(x-\mu)^2}{\sigma^2}\right)$
  $\mathcal N(\boldsymbol x|\boldsymbol\mu,\Sigma)=\frac{1}{\sqrt{(2\pi)^d|\Sigma|}}\exp\left(-\frac12(\boldsymbol x-\mu)^\top\Sigma^{-1}(\boldsymbol x-\mu)\right)$
  $X\sim\mathcal N(\boldsymbol\mu,\Sigma), Y=A+BX\Rightarrow Y\sim\mathcal N(A+B\boldsymbol\mu, B\Sigma B^\top)$
- Binom.: $f(k,n;p)=\mathbb P(X=k)=\binom nk p^k(1-p)^{n-k}$
- $\mathbb V[X]=\mathbb E[(X-\mathbb E[X])^2]=\mathbb E[X^2]-\mathbb E[X]^2$
  $\mathbb V[X+Y]=\mathbb V[X]+\mathbb V[Y]+2\mathbb{C}\mathrm{ov}(X,Y)$
- $\mathbb{C}\mathrm{ov}(X,Y)=\mathbb E[(X-\mathbb E[X])(Y-\mathbb E[Y])]$
  $=\mathbb E[XY]-\mathbb E[X]\mathbb E[Y]$
  $\mathbb{C}\mathrm{ov}(aX,bY)=ab\mathbb{C}\mathrm{ov}(X,Y)$

1.2 Calculus
- $\int uv'dx=uv-\int u'v\,dx$ • $\frac{\partial}{\partial x}\frac gh=\frac{g'h}{h^2}-\frac{gh'}{h^2}$
- $\frac{\partial}{\partial x}(b^\top Ax)=A^\top b$ • $\frac{\partial}{\partial x}(b^\top x)=\frac{\partial}{\partial x}(x^\top b)=b$
- $\frac{\partial}{\partial X}(c^\top X^\top b)=bc^\top$ • $\frac{\partial}{\partial X}(c^\top Xb)=cb^\top$
- $\frac{\partial}{\partial x}(x^\top Ax)=(A^\top+A)x \overset{A\text{ sym.}}{=} 2Ax$
- $\frac{\partial}{\partial X}Tr(X^\top A)=A$ • Tr. trick: $x^\top Ax \overset{\text{inner prod.}}{=}$
  $Tr(x^\top Ax)\overset{\text{cyclic permut.}}{=}Tr(xx^\top A)=Tr(Axx^\top)$
- $|X^{-1}|=|X|^{-1}$ • $\frac{\partial}{\partial X}\log|X|=X^{-\top}$ • $\frac{d}{dx}|x|=\frac{x}{|x|}$
- $\frac{\partial}{\partial x}\|x\|_2=\frac{\partial}{\partial x}(x^\top x)=2x$ • $\frac{\partial}{\partial x}\|x-b\|_2=\frac{x-b}{\|x-b\|_2}$
- $\frac{\partial}{\partial x}\|x\|_1=\mathrm{sgn}(x)$ $\mathrm{sgn}(x)\in\{\pm1\}^p$ is row-wise
- $\sigma(x)=\frac{1}{1+\exp(-x)}\Longrightarrow\nabla\sigma(x)=\sigma(x)(1-\sigma(x))$
- $\tanh x=\frac{2\sinh x}{2\cosh x}=\frac{e^x-e^{-x}}{e^x+e^{-x}}$ • $\nabla\tanh x=1-\tanh^2x$

2 Density Estimation
Bayesianism: Prior $P(\theta)$, likelihood $P(X|\theta)$, posterior $P(\theta|x_{1\dots n})$. **Bayes:** $P(\theta|X)=\frac{P(X|\theta)P(\theta)}{P(X)}$, $P(X)=\sum_\theta P(X|\theta_i)P(\theta_i)$
Frequentism: Param. model $P(Y|X,\theta)$, likelihood $P(X,Y|\theta)$. $\hat\theta_{MLE}=\arg\max_\theta P(X,Y|\theta)$.
Ex. Gaussian: $X_{i=1}^n\sim\mathcal N(\mu,\sigma^2)\Rightarrow\log\mathcal L_n(\mu,\sigma)=-n\log\sigma-\frac{nS^2}{2\sigma^2}-\frac{n(\bar X-\mu)^2}{2\sigma^2}\Rightarrow\hat\mu=\bar X,\hat\sigma=S$

2.1 Estimation - MLE Properties
Consistency: $\forall\epsilon>0,\mathbb P\{|\hat\theta_n-\theta^*|>\epsilon\}\xrightarrow{n\to\infty}0$
Equivariance: If $\hat\theta_n$ is MLE of θ, then $g(\hat\theta_n)$ is MLE of $g(\theta)$.
Asympt. normality: $\frac{1}{\sqrt n}(\hat\theta_n-\theta^*)\to\mathcal N(0,J^{-1}(\theta^*)I_n(\theta^*)J^{-1}(\theta^*))$
Asympt. efficiency: $\hat\theta_n$ minimises $\mathbb E[(\hat\theta_n-\theta^*)^2]$ as $n\to\infty$, i.e. $\mathbb E[(\hat\theta_n-\theta^*)^2]\overset{n\to\infty}{=}\frac{1}{I_n(\theta^*)}$ (Rao Cr.)
Among all consistent estimators $\hat\theta_n$ has *smallest variance*: $\lim_{n\to\infty}(\mathbb V[\hat\theta_n]I_n(\theta^*))^{-1}=1$

2.2 Rao Cramer inequality all $\mathbb E$ w.r.t. $P(x|\theta^*)$
Score func.: $\Lambda=\frac{\partial\log\mathbb P(x|\theta)}{\partial\theta}$, $\mathbb E[\Lambda]=0$
Fisher info.: $I_n(\theta)=\mathbb V[\Lambda]$
$J(\theta)=\mathbb E[\Lambda^2]=-\mathbb E\left[\frac{\partial^2\log\mathbb P(x|\theta)}{\partial\theta\partial\theta^\top}\right]=-\mathbb E\left[\frac{\partial\Lambda}{\partial\theta}\right]$
General bound: $\mathbb E[(\hat\theta_n-\theta^*)^2]\ge\frac{\left(1+\frac{\partial}{\partial\theta}b_{\hat\theta}\right)^2}{\mathbb E[\Lambda^2]}+b_{\hat\theta}^2$
Unbiased case: $\mathbb E[(\hat\theta_n-\theta^*)^2]=\mathbb V[\hat\theta_n]\ge\frac{1}{I_n(\theta^*)}$
Tradeoff: $\mathbb E[(\hat\theta_n-\theta^*)^2]=\mathbb V[\hat\theta_n]+\mathrm{bias}^2(\hat\theta_n)$
Bias: $\mathrm{bias}(\hat\theta_n)\equiv b_{\hat\theta}(\theta^*)=\mathbb E[\hat\theta_n]-\theta^*\overset{\text{unbiased}}{=}0$

3 (Linear) Regression model: $\hat{\mathbf y}=X\beta$
Assuming $X^\top X$ non-singular.
Bayesian view: $(Y|X,\beta)\sim\mathcal N(x^\top\beta,\sigma^2\mathbb I)$
Distrib. of estimator $\hat\beta_{LS}\sim\mathcal N(\beta,(X^\top X)^{-1}\sigma^2)$
Ridge: $\epsilon_{RSS}(\beta,\lambda)=(y-X^\top\beta)^\top(y-X^\top\beta)+\lambda\beta^\top\beta$
$\hat\beta=(X^\top X+\lambda\mathbb I)^{-1}X^\top y$, prior: $\beta\sim\mathcal N(0,\frac{\sigma^2}{\lambda}\mathbb I)$
(Ridge) Shrinkage: Decompose $X=UDV^\top$
$X\hat\beta=UD(D^2+\lambda\mathbb I)^{-1}DU^\top y=\sum_{j\le d}u_j\frac{d_j^2}{d_j^2+\lambda}u_j^\top y$
Lasso: $\hat\beta=\arg\min_\beta\sum_{i\le n}(y_i-x_i^\top\beta)^2+\lambda\|\beta\|_1$
(no closed form), prior: $p(\beta_i)=\frac{\lambda}{4\sigma^2}\exp(-|\beta_i|\frac{\lambda}{2\sigma^2})$
Bias-variance: $MSE=\mathbb E_{D,X,Y}[(\hat f(x)-Y)^2]$
$=(\mathbb E_{D,X}[\hat f(x)]-\mathbb E_Y[])]^2+\mathbb E_D[(\mathbb E_{D,X}[\hat f(x)]-\hat f(x))^2]+\mathbb E[(\mathbb E_Y[-]Y)^2]$
$=\mathrm{bias}^2+\mathrm{variance}+\mathrm{noise}$
Gauss-Markov Theorem: For any linear estimator $\tilde\theta=c^\top y=a^\top(\hat\beta+Dy)$ that is unbiased for $a^\top\beta$, it holds: $\mathbb V[a^\top\hat\beta]\le\mathbb V[c^\top y]$.
Among all linear **u**-estimators, $\hat\beta_{LS}$ minimises the gen. error!

3.1 3.1 Causality
Shortcut Learning: Model uses spurious correlation $X\leftrightarrow Y$ (e.g. grass ↔ cow)
$X_\alpha^\perp$: Part of X not causally influenced by α
Causal: $X_W^\perp$ and $X,W\&Y$ are causes of Y.
Anti-Causal: $X_W^\perp$ and $X,W\&Y$ are caused by Y.
W non-causal feature (e.g. background) influences X but not Y. **Counterfactually Inv.:** $f(X(w))=f(X(w'))$ for $w,w'\in W$. **Confounding:** Hidden U affects W,X. **Selection:** Hidden S filters on W,X. **Theorem For Counterfactual Invariance (CI):** Anti-causal: $f(X)\perp W|Y$ Causal (no selection): $f(X)\perp W$ Causal (no confounding): $Y\perp X|X^{\perp W},W$ and $f(X)\perp W|Y$.

4 Uncertainty Quantification
4.1 Bayesian Linear Regression
Model: $y=X^\top\beta+\epsilon$, with $\epsilon\sim\mathcal N(\epsilon|0,\sigma^2\mathbb I)$
Likelihood: $P(y|X,\beta,\sigma)=\mathcal N(y|X^\top\beta,\sigma^2\mathbb I)$
Prior: $P(\beta|\Lambda)=\mathcal N_d(\beta|0,\Lambda^{-1})$
(Ridge regr. if $\Lambda=\lambda\mathbb I$ and $\sigma=1$)
Posterior: $P(\beta|X,y,\Lambda)=\mathcal N(\beta|\mu_\beta,\Sigma_\beta)$
with $\mu_\beta=(X^\top X+\sigma^2\Lambda)^{-1}X^\top y$
and $\Sigma_\beta=\sigma^2(X^\top X+\sigma^2\Lambda)^{-1}$
$\Longrightarrow y\sim\mathcal N(y|0,X\Lambda^{-1}X^\top+\sigma^2\mathbb I)$ using $\mathbb E_{\beta,\epsilon}[\cdot]$
kernel $k(x_i,x_j):=x_i^\top\Lambda^{-1}x_j$

4.2 Gaussian Process
$y\sim\mathcal N(y|m(X),K(X,X)+\sigma^2\mathbb I)$
$\begin{bmatrix}y\\y_{n+1}\end{bmatrix}\sim\mathcal N\left(\begin{bmatrix}y\\y_{n+1}\end{bmatrix}\middle|\begin{bmatrix}m(X)\\m(x_{n+1})\end{bmatrix},\begin{bmatrix}C_n&k\\k^\top&c\end{bmatrix}\right)$
$p(y_{n+1}|x_{n+1},X,y)=\mathcal N(y_{n+1}|\mu_{n+1},\sigma_{n+1}^2)$
with $\mu_{n+1}=m(x_{n+1})+k^\top C_n^{-1}(y-m(X))$
and $\sigma_{n+1}^2=c-k^\top C_n^{-1}k$
where $K=k(X,X)$, $k=k(x_{n+1},X)$,
$C_n=K+\sigma^2\mathbb I$, $c=k(x_{n+1},x_{n+1})+\sigma^2$

4.3 Kernels scalar product $K_{ij}=k(x_i,x_j)$
Valid kernel: must be **symmetric** and **p.s.d.** ($x^\top Kx\ge0\,\forall x$ *or* pos. eigenvalues *or* pos. principal minors). Must have a (pot. ∞-dim.) **feature vector** ϕ s.t. $k(x,x')=\phi(x)^\top\phi(x')$.
Common kernels:
Linear: $x^\top x'$
Polynomial: $(x^\top x'+1)^p$, $p\in\mathbb N$
RBF (Gaussian): $\exp(-\|x-x'\|_2^2/h^2)$
Sigmoid: $\tanh(\kappa\cdot x^\top x'-b)$
Kernel construction: • k_1+k_2 • $c\cdot k_1, c>0$
• $k_1\cdot k_2$ • $f(x)k_1(x,x')f(x')$
• $k(\phi(x),\phi(x'))$ with $\phi:\mathcal X\to\mathbb R^d$
• $g(k_1)$ with g: exp. *or* polyn. w/ all pos. coeff.

4.4 Active Learning $I(X;Y)=H(X)-H(X|Y)$
Mutual Info: $I(X;Y)=\sum_{x,y}p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$
Transd.-L.: $x_n=\arg\max_{x\in S}I(f_{x\in A};y_x|D_{n-1})$
→ For $f\sim GP(\mu,k)$: $I=\frac12\log\frac{\mathrm{Var}(y_x|D_{n-1})}{\mathrm{Var}(y_x|f_{x\in A},D_{n-1})}$
Safe BO: GP s_{f^*},g^* observe $y_i=f^*(x_i), z_i=g^*(x_i)$
95% conf.: $[l_n^f,u_n^f],[l_n^g,u_n^g]$
$S_n=\{x:l_n^g(x)\ge0\}$, $\hat S_n=\{x:u_n^g(x)\ge0\}$
$A_n=\{x\in\hat S_n:u_n^f(x)\ge\max_{x'\in S_n}l_n^f(x')\}$
Now ITL on sample space S_n, target space A_n.
Batch Active Learning: $B_\delta(x)=\{x':\|x-x'\|\le\delta\}$
$\Rightarrow\arg\max_{L\subseteq X,|L|=b}\Pr(\cup_{x\in L}B_\delta(x))$
$\Rightarrow\arg\max_{L\subseteq X,|L|=b}\frac{1}{|X|}|\{x':\|x'-x\|\le\delta,x\in L\}|$
NP-hard Problem, greedy algo (ProbCover):
$L=\emptyset, E=\{(x,x'):\|x-x'\|\le\delta\}$ For $i=1,\dots b$:
Add $x_i=\arg\max_{x\in X}|\{x':(x,x')\in E,x'\in X\}|$
Update $L\leftarrow L\cup\{x_i\}, E\leftarrow E\setminus(\{x_i\}\times(B_\delta(x_i)\cap X))$

5 Support Vector Machine (SVM)
Primal (soft margin): $\min_{w,w_0,\xi}\frac12\|w\|^2+C\sum_i\xi_i$
s.t. $y_i(w^\top x_i+w_0)\ge1-\xi_i$ and $\xi_i\ge0$
↪ intractable if $\varphi(x_i)$ instead of x_i
↪ $\xi_i=0$ means x_i was not neglected
Dual: $\max_\alpha\sum_i\alpha_i-\frac12\sum_{(i,j)}\alpha_i\alpha_jy_iy_jx_i^\top x_j$
s.t. $0\le\alpha_i\le C$; $\sum_i\alpha_iy_i=0$
↪ solve α via quadratic optimisation
Optimal hyperplane: $w^*=\sum_i\alpha_i^*y_ix_i$
↪ $\alpha_i^*\ne0$ only for *support vectors*
Optimal slack: $\xi_i^*=\max(0,1-y_i(w^{*\top}x_i+w_0^*))$

5.1 Structural SVMs
$\min_{w,\xi}\frac12\|w\|^2+\frac Cn\sum_{i\le n}\xi_i$ s.t. $\xi_i\ge0$ and $\forall y'\ne y_i$:
$w^\top\Psi(x_i,y_i)\ge\Delta(y_i,y')+\underbrace{w^\top\Psi(x_i,y')}_{\hookrightarrow\text{ mislabelings}}-\xi_i-\epsilon$
Ψ: *joint-feature map*; Δ: loss / class dissimilarity func.; $w^\top\Psi(x,y)$: compatibility score btw. x and y; ϵ: tolerance / universal slack variable.
Prediction: $c(x)=\arg\max_y w^\top\Psi(x,y)$
Note: For optimal w^*,ξ^*, emp. risk$(w^*)\le\frac1n\sum_i\xi_i^*$
Training: Start without any constraints. In each iteration, add for each (x_i,y_i) the constraint with $y'\ne y_i$ that is the "most violated" and solve again with quadr. optimisation.

6 Ensemble Methods $\hat f(x)\triangleq\frac1B\sum_{i\le B}\hat f_i(x)$
$\mathrm{bias}[\hat f(x)]=\frac1B\sum\mathrm{bias}[\hat f_i(x)]$
$\mathbb V[\hat f]=\frac{1}{B^2}\sum\mathbb V_D[\hat f_i]+\frac{1}{B^2}\sum\sum_{i,j}\mathbb{C}\mathrm{ov}(()\hat f_i,\hat f_j)\approx\frac{\sigma^2}{B}$

6.1 Bagging (Bootstrap aggregation)
1. Draw M bootstrap sets $Z_1^*,\dots,Z_M^*$
2. Train M base models $b^{(1)},\dots,b^{(M)}$
3. **Aggregate:** $b^{(M)}(x)=\begin{cases}\frac1M\sum_{t\le M}b^{(t)}(x)&\text{regr.}\\\mathrm{sign}(\sum_t b^{(t)}(x))&\text{class.}\end{cases}$

Why it works: Small *variance* (weak learners)
Random Forest: At each splitting step, u.a.r. choose m of p features & split only one (best) feature. → reduce *correlation* between trees.

6.2 Boosting

Sequentially train weak learners on all data, but ↗ weight of misclass. samples (↘ bias).
AdaBoost: Stat. learning (*forward stagewise additive modeling*) with **exp. loss**, trains max-margin ($=y_i\bar{b}(\boldsymbol{x}_i)$), self-avg. and interpolating (↘ overfitting) classifiers.

[Init]: $w_i \leftarrow 1/n\ \forall i\le n$ for $b=1\ldots B$:
[Train]: Fit classifier $c_b(x)$ using weights w_i
[Eval]: $\epsilon_b=\frac{\sum_{i=1}^n w_i\mathbb{I}\{c_b(x_i)\neq y_i\}}{\sum_{i=1}^n w_i}$ // error of c_b
[Update]: $\alpha_b=\log(\frac{1-\epsilon_b}{\epsilon_b})$ // always >0, final weight of c_b
[Reweight]: $w_i\leftarrow w_i\exp(\alpha_b\mathbb{I}\{c_b(x_i)\neq y_i\})$ normalize!
Return $\hat{c}_{\text{ADA}}(x)=\text{sign}(\sum_{b=1}^B\alpha_b c_b(x))$

7 Deep Learning

Sigmoid: $\sigma(x)=\frac{1}{1+\exp(-x)}=\frac{e^x}{e^x+1}$
$\sigma'(x)=\sigma(x)(1-\sigma(x))=\sigma(x)\sigma(-x)$
Softmax: $y_i\propto\exp(\beta z_i)$
Backpropagation: Gradient: $\frac{\partial\ell}{\partial w_{jk}}=\delta_j^{(l)}v_k^{(l-1)}$

7.1 Variational Autoencoders

Def: $\xrightarrow[\text{prior}]{p_{\theta'}(z)}\mathcal{Z}\xrightarrow[\text{likelihood}]{\text{dec}_\theta(z)=p_\theta(x|z)}\mathcal{X}\xrightarrow[\text{approx. posterior}]{\text{enc}_\phi(x)=q_\phi(z|x)}\mathcal{Z}$
sample/obs. $\mathcal{X}_i$ from latent representation $\mathcal{Z}$
Train: $\max_{\theta',\theta,\phi}\underbrace{\sum_i\log p_{\theta',\theta}(x_i)}_{(*)\text{ indep. of }Z}$

$(*)=\mathbb{E}_{Z\sim q_\phi(\cdot|x_i)}\left[\log\left(\frac{p_{\theta',\theta}(x_i,Z)}{p_{\theta',\theta}(Z|x_i)}\frac{q_\phi(Z|x_i)}{q_\phi(Z|x_i)}\right)\right]$

$\overbrace{\mathcal{L}(x_i,\theta,\phi)\equiv\textbf{ELBO}=\text{Infomax}-\text{Regularisation term}}$

$=\mathbb{E}[\log p_\theta(x_i|Z)]-D_{\text{KL}}\left(q_\phi(\cdot|x_i)\|p_{\theta'}(\cdot)\right)$
$+D_{\text{KL}}\left(q_\phi(\cdot|x_i)\|p_{\theta',\theta}(\cdot|x_i)\right)\ge\mathcal{L}(x_i,\theta,\phi)$

Train: $\theta^*,\phi^*=\arg\max_{\theta,\phi}\mathcal{L}(x_i,\theta,\phi)$

Requirements for good representation:
- **informative:** $\theta^*=\arg\max_\theta I(X;Z)$
  $=\arg\max_\theta\mathbb{E}_{X,Z}[\log p(X|Z)]-const_{\text{w.r.t. }\theta}$
  $\approx\arg\max_\theta\sum_i\mathbb{E}_{Z|X}[\log p(x_i|Z)]$
- **disentangled:** components in $\mathcal{Z}$ associated with distinct feature in $\mathcal{X}$ (see D_{KL} in ELBO).
- **robust:** noise in $\mathcal{Z}$ doesn't substantially affect $\mathcal{X}$ (and vice versa). → choice of approx. post.!

7.2 Transformers

Attention: Query: $Q=EW^Q$, Key: $K=EW^K$, Value: $V=EW^V$, embedding E. $A=\text{softmax}(\frac{QK^\top}{\sqrt{d}})V$, d: E-dim.

- **Self-Attention:** Q,K,V from same sequence
- **Cross-Att.:** Q from target, K,V from source.
- **Multi-headed:** Multiple weights, Stack A_i's.
- **Masked Self-Attention:** Future tokens masked ($-\infty$ in upper triangle of $P=QK^\top$)

Pos. Encoding: $p_{i,2j}=\sin(\alpha^{-2j}i)$, $p_{i,2j+1}=\cos(\alpha^{-2j}i)$, for $j\le d/2$, $\alpha=10^{4/d}$.

7.3 Stable Diffusion

Forward Process: For $t\le T$, $\beta_t>0$, let $\alpha_t=1-\beta_t$, $\bar{\alpha}_t=\prod_{s\le t}\alpha_s$
$q(x_t|x_{t-1})=\mathcal{N}(x_t|\sqrt{1-\beta_t}x_{t-1},\beta_tI)$
$q(x_t|x_0)=\mathcal{N}(x_t|\sqrt{\bar{\alpha}_t}x_0,(1-\bar{\alpha}_t)I)$
Reverse Process: UNet predicts noise $\epsilon_\theta(x_t,t)$ added to form x_t from x_0. Then reconstruct x_{t-1}:
$x_{t-1}=\frac{1}{\sqrt{\alpha_t}}\left(x_t-\frac{\sqrt{1-\alpha_t}}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t,t)\right)+\sigma_tz,z\sim\mathcal{N}(0,I)$

7.4 Reinforcement Learning

Rew.-to-go: $G_t=\sum_{k=0}^\infty\gamma^kr_{t+k}=r_t+\gamma G_{t+1}$. **Markov Property:** $P(s_{t+1}|s_{0:t},a_{0:t})=P(s_{t+1}|s_t,a_t)$. **Values:** $V^\pi(s)=\mathbb{E}_\pi[G_t|s_t=s]$, $Q^\pi(s,a)=\mathbb{E}_\pi[G_t|s_t=s,a_t=a]$. **Bellman Exp.:** $V^\pi(s)=\sum_{a,s'}\pi(a|s)p(s'|s,a)(r+\gamma V^\pi(s'))$. **" Opt.:** $Q^*(s,a)=\sum_{s'}p(s'|s,a)(r+\gamma\max_{a'}Q^*(s',a'))$. **Q-Learning:** $Q(s,a)\leftarrow Q(s,a)+\alpha[r+\gamma\max_{a'}Q(s',a')-Q(s,a)]$ (Off-policy). **DQN:** MLP $Q_\theta\approx Q^*$. Loss $L(\theta)=(r+\gamma\max_{a'}Q_{\theta^-}(s',a')-Q_\theta(s,a))^2$. Uses **Replay Buffer** (decorrelates data) and **Target Network** θ^- (stablises targets). **Policy Gradient:** $\nabla_\theta J(\theta)=\mathbb{E}[\sum_t\nabla_\theta\log\pi_\theta(a_t|s_t)G_t]$. **REINFORCE:** 1) Initialize env. 2) For each epoch 2.1) Initialize buffer. 2.2) For each episode: collect data using π_θ, compute G_t, store in buffer. 2.3) Compute $\nabla_\theta L_{PG}(\theta)=-\frac{1}{|D|}\sum_D\sum_t\nabla_\theta\log\pi_\theta(a_t|s_t)G_t$ using buffer. 2.4) Update θ using $\nabla_\theta L_{PG}(\theta)$.
. **Actor-Critic:** Actor π_θ and Critic V_ϕ. Advantage $A_t=G_t-V_\phi(s_t)$. **GAE:** $A_t^{GAE}=\sum_{l=0}^{T-t-1}(\gamma\lambda)^l\delta_{t+l}$ where $\delta_t=r_t+\gamma V_\phi(s_{t+1})-V_\phi(s_t)$. **AC+GAE:** 1) Initialize env., actor, critc 2-2.2) *Same* 2.3) For each critic update compute critic loss and optimize it. 2.4) Compute values for each tstamp using critic. 2.5) Compute GAE. 2.6) Compute actor loss and optimize it (policy). **PPO:** $L_{CLIP}(\theta)=-\mathbb{E}[\min(w_tA_t,\text{clip}(w_t,1-\epsilon,1+\epsilon)A_t)]$, $w_t(\theta)=\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$. 1-2.2) *Same as AC+GAE* ~~2.3)~~ 2.4-2.5) *Same as AC+GAE* 2.6) For each update 2.6.1) Compute critic loss and update it 2.6.2) Same with actor. **Entropy Reg.:** $L_{total}=L_{policy}+c_VL_V-c_HH(\pi_\theta)$ (prevents premature collapse).

8 Clustering

***k*-means:** $\arg\min_\theta\sum_{i\le n}\|\boldsymbol{x}_i-\theta_{c(\boldsymbol{x}_i)}\|^2$

8.1 Mixture Models

Assume: $\boldsymbol{x}\sim p(\boldsymbol{x}|\pi_{1\ldots k},\theta_{1\ldots k})=\sum_{c\le k}\pi_cp(\boldsymbol{x}|\boldsymbol{\theta}_c)$
Find: $\hat{\theta}=\arg\max_\theta p(\mathcal{X}|\pi,\theta)=\prod_x p(x|\pi,\theta)$
Gaussian Mixtures: → $p(\boldsymbol{x}|\boldsymbol{\theta}_c)=p(\boldsymbol{x}|\boldsymbol{\mu},\Sigma)$
Introduce *latent indicator variables* for mode assignments $M_{xc}\in\{0,1\}$. Then, the **log-likelihood:**
$L(\mathcal{X},\boldsymbol{M}|\boldsymbol{\theta})=\sum_x\sum_{c\le k}M_{xc}\log(\pi_cp(\boldsymbol{x}|\boldsymbol{\theta}_c))$

8.1.1 EM-Algorithm for Gaussian Mixtures

E-step: Calculate
$Q(\boldsymbol{\theta};\boldsymbol{\theta}^{(t)})=\mathbb{E}_{\boldsymbol{M}|\mathcal{X},\boldsymbol{\theta}^{(t)}}[L(\mathcal{X},\boldsymbol{M}|\boldsymbol{\theta})]=\ldots$
$=\sum_x\sum_{c\le k}\left(\underbrace{\mathbb{E}_{\boldsymbol{M}|\mathcal{X},\boldsymbol{\theta}^{(t)}}[M_{xc}]}\cdot\log\pi_cp(\boldsymbol{x}|\boldsymbol{\theta}_c)\right)$
where $\gamma_{xc}=\frac{p(x|c,\theta^{(t)})p(c|\theta^{(t)})}{p(x|\theta^{(t)})}$, $\sum_{c\le k}\gamma_{xc}=1$
M-step: $\boldsymbol{\theta}^{(t+1)}\in\arg\max_\theta Q(\boldsymbol{\theta};\boldsymbol{\theta}^{(t)})$
s.t. $\sum_c\pi_c=1$. Solve via Lagrangian, yields
$\pi_c=\frac{1}{|\mathcal{X}|}\sum_x\gamma_{xc}$, $\boldsymbol{\mu}_c=\frac{\sum_x\gamma_{xc}\boldsymbol{x}}{\sum_x\gamma_{xc}}$, $\sigma_c^2=\frac{\sum_x\gamma_{xc}(x-\mu_c)^2}{\sum_x\gamma_{xc}}$

8.2 Non-parametric Bayesian Methods

$\text{Dir}(\boldsymbol{x}|\boldsymbol{\alpha})=\frac{1}{B(\boldsymbol{\alpha})}\prod_{k=1}^nx_k^{\alpha_k-1}$, $B(\boldsymbol{\alpha})=\frac{\prod_{k=1}^n\Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^n\alpha_k)}$
Rewrite **Finite mixture models:**
$p(x)=\sum_{k=1}^K\pi_kp(x|\theta_k)=\int p(x|\theta)G(\theta)\mathrm{d}\theta$
where $G(\theta)=\sum_{k=1}^\infty\pi_k\delta_{\theta_k}(\theta)$ ← discrete distr.

8.2.1 Finite GMM

1. Cluster centers: $\mu_k\sim\mathcal{N}(\mu_0,\sigma_0)$
2. Prob's of clusters: $\pi_{1\ldots K}\sim\text{Dir}(\alpha_{1\ldots K})$
3. Cluster assignments: $z_i\sim\text{Categorical}(\pi_{1\ldots K})$
4. Coordinates of data: $x_i\sim\mathcal{N}(\mu_{z_i},\sigma_{z_i})$

8.2.2 DP Mixture Model (DP-GMM)

1. Cluster centers: $\mu_k\sim\mathcal{N}(\mu_0,\sigma_0)$, $k=1,2,\ldots$
2. Prob's of clusters: $\pi=(\pi_1,\pi_2,\ldots)\sim\text{GEM}(\alpha)$
3. Cluster assignments: $z_i\sim\text{Categorical}(\pi)$
4. Coordinates of data: $x_i\sim\mathcal{N}(\mu_{z_i},\sigma)$, $i=1\ldots N$

Fitting a DP-MM: **Collapsed Gibbs sampler**
$p(z_i=k|\boldsymbol{z}_{-i},\boldsymbol{x},\alpha,\boldsymbol{\mu})\propto\text{Prior}\times\text{Likelihood}$
$\propto\begin{cases}\frac{|\boldsymbol{x}_{-i,k}|}{\alpha+N-1}p(x_i|\boldsymbol{x}_{-i,k},\boldsymbol{\mu}) & \text{for existing } k\\ \frac{\alpha}{\alpha+N-1}p(x_i|\boldsymbol{\mu}) & \text{otw.}\end{cases}$
$\boldsymbol{x}_{-i,c}:=\{x_j|z_j=c,j\neq i\}$ data assigned to clust. c

9 PAC Learning

Want: Distribution indep. error guarantees!
Expec./**Gener. error:** $\mathcal{R}(\hat{c}_n)=\boldsymbol{P}_{X,Y}(\hat{c}_n(x)\neq c(x))$
Empirical error: $\hat{\mathcal{R}}_n(\hat{c}_n)=\frac{1}{n}\sum_{i=1}^n\mathbf{1}_{\{\hat{c}_n(x_i)\neq y_i\}}$
PAC learnable: $\mathcal{A}$ can learn a concept class $\mathcal{C}$ from $\mathcal{H}$ if, given a suff. large sample, it outputs a hypothesis that generalizes well w/ high prob.

(1) $0<\epsilon<1/2$, $0<\delta<1/2$, (2) $\boldsymbol{P}_{X,Y}$ on $\mathcal{X}\times\{0,1\}$:
If $n\ge poly(1/\epsilon,1/\delta,dim(\mathcal{X}))$,
Then $\boldsymbol{P}_{X,Y}\left(\mathcal{R}(\hat{c}_n)-\inf_{c\in\mathcal{C}}\mathcal{R}(c)\le\epsilon\right)\ge1-\delta$.

If $\mathcal{A}$ runs in time polynomial in $1/\epsilon$ and $1/\delta$, we say that $\mathcal{C}$ is **efficiently PAC learnable**.

9.1 VC Inequality $\boldsymbol{P}(\cdots\ge\epsilon)\le\ldots\le\delta$

Select ERM: $\hat{c}_n^*=\arg\min_{c\in\mathcal{C}}\hat{\mathcal{R}}_n(c)$
Under uniform convergence:
$\boldsymbol{P}\left(\mathcal{R}(\hat{c}_n^*)-\inf_{c\in\mathcal{C}}\mathcal{R}(c)>\epsilon\right)\le\boldsymbol{P}\left(\sup_{c\in\mathcal{C}}\left|\hat{\mathcal{R}}_n(c)-\mathcal{R}(c)\right|>\frac{\epsilon}{2}\right)$
- $|\mathcal{C}|$ **Finite:** $\boldsymbol{P}(\sup|\cdots|>\epsilon)\le2|\mathcal{C}|\exp(-2n\epsilon^2)$
- $|\mathcal{C}|$ **Unbounded:** $\boldsymbol{P}(\cdots)\le9n^{\text{VC}_\mathcal{C}}\exp(-\frac{n\epsilon^2}{32})$

9.2 Rectangle Learning

$\boldsymbol{P}((\hat{c}_n^*>\epsilon)\le|\mathcal{C}|\cdot(1-\epsilon)^n\le|\mathcal{C}|\cdot\exp(-n\epsilon)<\delta$
Union bound: $\boldsymbol{P}(\bigcup_iT_i)\le\sum_i\boldsymbol{P}(T_i)$

10 Appendix

Complete the square: If $p(\boldsymbol{x})\propto\exp(-\frac{1}{2}\boldsymbol{x}^\top\boldsymbol{A}\boldsymbol{x}+\boldsymbol{x}^\top\boldsymbol{b})$, then $p(\boldsymbol{x})=\mathcal{N}(\boldsymbol{x}|\boldsymbol{A}^{-1}\boldsymbol{b},\boldsymbol{A}^{-1})$
Constrained optimisation:
primal: $\min_xf(\boldsymbol{x})$ s.t. $g_i(\boldsymbol{x})=0$; $h_j(\boldsymbol{x})\le0$
Lagrangian: with each $\alpha_j\ge0$
$\mathcal{L}(\boldsymbol{x},\lambda,\alpha)=f(\boldsymbol{x})+\sum_i\lambda_ig_i(\boldsymbol{x})+\sum_j\alpha_jh_j(\boldsymbol{x})$
Solve: $\frac{\partial\mathcal{L}}{\partial\boldsymbol{x}}=0$; $g_i(\boldsymbol{x})=0$; $\alpha_j\ge0$; $h_j(\boldsymbol{x})\le0$
If **Slater's cond.** holds, $\exists\boldsymbol{x}:g_i(\boldsymbol{x})=0,h_j(\boldsymbol{x})<0$, then we can solve the *dual* instead:
$\max_{\lambda,\alpha}\{\min_x\mathcal{L}(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\alpha})\}$ s.t. $\alpha_j\ge0$
Solve: $\frac{\partial\mathcal{L}}{\partial\boldsymbol{x}}=0$; $\frac{\partial\mathcal{L}}{\partial\lambda}=0$; $\alpha_jh_j(\boldsymbol{x})=0$; $\alpha_j\ge0$
Metrics: $acc=\frac{TP+TN}{n}$ $prec=\frac{TP}{TP+FP}$ $FPR=\frac{FP}{FP+TN}$
$Recall/TPR=\frac{TP}{TP+FN}$ $balanced acc=\frac{1}{n}\sum_iTPR_i$
$F1-score=\frac{2TP}{2TP+FP+FN}$ $ROC=FPR/TPR$
Conditional Gaussians:
$P_{X,Y}=\begin{bmatrix}\mathbf{X}\\\mathbf{Y}\end{bmatrix}\sim\mathcal{N}\left(\begin{bmatrix}\mu_X\\\mu_Y\end{bmatrix},\begin{bmatrix}\Sigma_{XX}&\Sigma_{XY}\\\Sigma_{YX}&\Sigma_{YY}\end{bmatrix}\right)$, Σ_{ij} p.s.d.

$\Longrightarrow Y|X \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}, \tilde{\Sigma})$, where $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu}_Y + \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}(X - \boldsymbol{\mu}_X)$, $\tilde{\Sigma} = \boldsymbol{\Sigma}_{YY} - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY})$